

Cascade Directionality Detection via MIL Training Dynamics

Session Experiment Report

Cytokine-MIL Project

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Background and Motivation

Setting

An AB-MIL model was trained to classify which cytokine was applied to a pseudo-tube of PBMCs (91 classes: 90 cytokines + PBS). Each pseudo-tube contains ≈ 480 cells stratified by 18 cell types across 12 donors (Donor2 and Donor3 held out for validation).

Training proceeds in three stages:

1. **Stage 1:** Encoder pre-training with cell-type classification supervision.
2. **Stage 2:** MIL fine-tuning with frozen encoder (20 epochs, $\text{LR} = 10^{-3}$).
3. **Stage 3:** Full fine-tuning, encoder unfrozen (100 epochs).

During Stage 3, per-cell-type centroids were logged at every epoch for all cytokines and training donors. Experiments across **10 random seeds** were run to assess robustness.

Known Cascade Pairs (Benchmark)

The PBS-RC static geometry method (preceding this session) recovers all 11 known cascade pairs with high sensitivity, but calls *every* pair as “shared” (both $A \rightarrow B$ and $B \rightarrow A$ significant). The open problem that this session addresses is **directionality**: which cytokine is upstream?

Table 1: Known cytokine cascade pairs used as benchmark throughout this session.

Source (A)	Target (B)	Known relay cell types
IL-12	IFN- γ	NK, NKT, CD8 T
IL-1-beta	IL-6	Monocytes, macrophages
IL-2	IL-15	T cells, NK
IL-33	IL-13	ILC2, mast cells
IL-18	IFN- γ	NK, T cells
IL-21	IL-10	B cells, plasma cells
TNF-alpha	IL-6	Stromal, monocytes
IFN-alpha1	IFN- γ	NK, T cells
IL-10	IL-6	Monocytes (regulatory)
IL-4	IL-13	Th2, ILC2
IL-27	IFN- γ	NK, T cells

Model Performance

The trained model achieves $P_{\text{correct}} \approx 0.039$ on training data ($3.5\times$ above chance for 91 classes). This relatively low accuracy is an important caveat for interpreting all experiments in this session.

Experiment 1: Attention-Weighted Centroid PCA Trajectories

Hypothesis

If $A \rightarrow B$ via relay cell type T, the model should learn to represent T cells in A-tubes as B-like. Under fine-tuning, the attention-weighted centroid of T cells in A-tubes ($\tilde{\mu}_{A,T}^{\text{attn}}$) should drift *toward* the centroid of T cells in B-tubes in the latent space. Projecting both cytokines’ centroids into a shared PCA 2D subspace (fitted on all centroids from source and target across all epochs) should reveal this convergence as a trajectory plot.

Method

For each seed and each cell type T :

1. Compute attention-weighted centroid: $\tilde{\mu}_{A,T}(t) = \text{mean}_{i:\tau(i)=T} a_i(t) \cdot \tilde{h}_i(t)$, averaged over training donors.
2. Fit PCA on all centroids of source and target (all cell types, all epochs) for that seed.
3. Project source and target centroids into the 2D cascade-relevant subspace.
4. Track the trajectory over training epochs.

Analysis: 10 seeds $\times$ 18 cell types, focused on IL-12 $\rightarrow$ IFN- γ .

Results and Failure Mode

Per-seed visualizations showed inconsistent patterns across seeds. The key failure mode was identified: **the attention-weighted centroid is contaminated by attention magnitude**. For low-attention cell types (e.g., B cells), $a_i \approx 0$ for all cells, so the weighted centroid collapses to $\approx \mathbf{0}$ for *both* cytokines, producing spurious convergence. For high-attention relay cells, the model pushes their embeddings to be *maximally discriminative* (far from each other), not similar — producing apparent divergence.

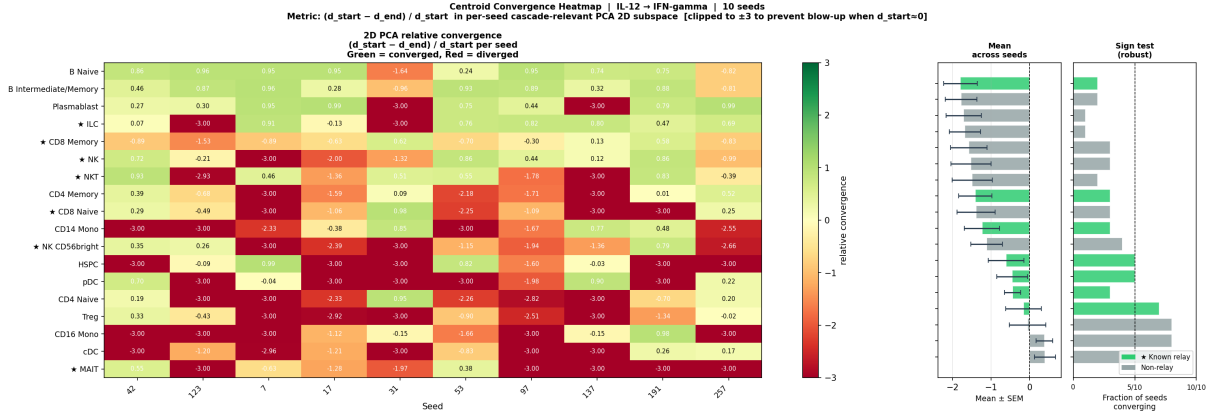


Figure 1: **PCA distance convergence heatmap** (IL-12 $\rightarrow$ IFN- γ , 10 seeds $\times$ 18 cell types). Metric: $(d_{\text{start}} - d_{\text{end}}) / d_{\text{start}}$ in the per-seed cascade-relevant 2D PCA subspace; clipped to ± 3 . Green = converged (source and target centroids moved closer). Red = diverged. **Sign test (right panel):** B Naive and B Intermediate converge in $>80\%$ of seeds — biologically incorrect (B cells are not relay cells). NK, NKT converge in only $\approx 40\text{--}50\%$ of seeds (near chance), despite being the known relays. The metric is fundamentally unreliable.

Conclusion

Null result. The attention-weighted centroid convergence metric does not reliably identify relay cells or cascade direction. The attention weighting itself is the confound: low-attention cells produce trivial convergence; relay cells (high attention) diverge because the model makes them discriminative.

Experiment 2: Epoch-Dependent PBS-RC Directional Bias Asymmetry

Hypothesis

If $A \rightarrow B$ via relay T , the attention-weighted centroid of T cells in A -tubes should deviate toward B 's global embedding, more than the symmetric reverse signal. Define:

$$b_{\text{fwd}}(A \rightarrow B, T, t) = (\tilde{\mu}_{A,T}(t) - \tilde{\mu}_A(t)) \cdot \hat{u}_{A \rightarrow B}(t), \quad (1)$$

$$b_{\text{fwd}}(B \rightarrow A, T, t) = (\tilde{\mu}_{B,T}(t) - \tilde{\mu}_B(t)) \cdot \hat{u}_{B \rightarrow A}(t), \quad (2)$$

where $\tilde{\mu}_A(t)$ is A 's global centroid (mean over all cell types) and $\hat{u}_{A \rightarrow B}(t)$ is the unit direction from A to B at epoch t . The asymmetry $\text{Asym}(A \rightarrow B, T, t) = b_{\text{fwd}}(A \rightarrow B) - b_{\text{fwd}}(B \rightarrow A)$ should be **positive and growing** for cascade pairs.

Method

Computed from existing `dynamics_stage3.pkl` files (no new training). For each of the 11 known pairs, computed b_{fwd} at every logged epoch across all training donors and 10 seeds. Summarised by mean asymmetry over epochs and cell types.

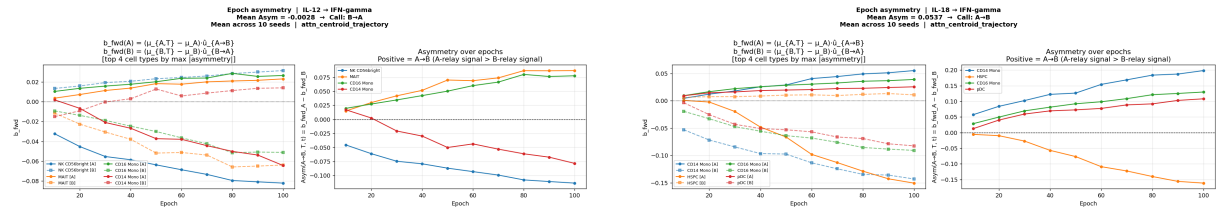
Results

Table 2: Epoch asymmetry summary. Mean Asym = $\overline{\text{Asym}(A \rightarrow B, T, t)}$ averaged over all cell types and epochs. Call: $A \rightarrow B$ if Mean Asym > 0 . $\checkmark$ = matches known direction; $\times$ = does not match.

A	B	Mean Asym	Best relay T	Call	
IL-12	IFN- γ	-0.0028	MAIT	B $\rightarrow$ A	$\times$
IL-2	IL-15	+0.0443	cDC	A $\rightarrow$ B	$\checkmark$
IL-33	IL-13	+0.0102	cDC	A $\rightarrow$ B	$\checkmark$
IL-18	IFN- γ	+0.0537	CD14 Mono	A $\rightarrow$ B	$\checkmark$
IL-21	IL-10	-0.0040	NKT	B $\rightarrow$ A	?
IL-10	IL-6	-0.0210	ILC	B $\rightarrow$ A	$\times$
IL-4	IL-13	+0.0223	Plasmablast	A $\rightarrow$ B	$\checkmark$
IL-27	IFN- γ	-0.0064	CD14 Mono	B $\rightarrow$ A	$\times$

3 pairs absent from trajectory data (IL-1-beta, TNF-alpha, IFN-alpha1).

Score: 4/8 correct (50% = chance).



(a) IL-12 $\rightarrow$ IFN- γ : essentially flat asymmetry across all 100 epochs. MAIT identified as best relay (biologically incorrect; NK cells are the known relay). Mean Asym = $-0.003 \approx 0$.

(b) IL-18 $\rightarrow$ IFN- γ : positive and growing asymmetry (CD14 Mono, CD4 Naive). Mean Asym = $+0.054$. This is what a genuine directional signal would look like, but the relay cell type is biologically unexpected.

Figure 2: Epoch-dependent directional bias asymmetry for two representative pairs. **Left panel of each figure:** b_{fwd} for A -tubes (solid) and B -tubes (dashed) per cell type over training epochs. **Right panel:** $\text{Asym}(A \rightarrow B, T, t)$; positive = $A \rightarrow B$ signal. Mean across 10 seeds.

Conclusion

Null result for cascade directionality ($4/8 = \text{chance}$). The canonical IL-12 $\rightarrow$ IFN- γ pair shows zero signal. Asymmetry magnitudes are uniformly small ($|\text{Asym}| < 0.06$). The same root cause as Experiment 1 applies: the attention-weighted centroid confounds geometry with attention magnitude.

Experiment 3: Confusion Matrix Asymmetry

Hypothesis

If $A \rightarrow B$ via relay cells, A-tubes contain cells that transcriptionally resemble B (they are producing B in response to A). The trained 91-class classifier should therefore assign higher softmax probability to class B when evaluating A-tubes than it assigns to class A when evaluating B-tubes:

$$P(B \mid A\text{-tube}) > P(A \mid B\text{-tube}) \implies A \rightarrow B.$$

This defines the asymmetry: $\text{Asym}(A \rightarrow B) = C[A, B] - C[B, A]$, where $C[i, j] = \overline{P(j \mid \text{tube of class } i)}$.

Method

For each of the 10 seeds, loaded `model_stage3.pt`, ran a forward pass on all 9,100 training tubes, and accumulated the mean-softmax 91×91 confusion matrix. Averaged across seeds and computed element-wise asymmetry.

Results

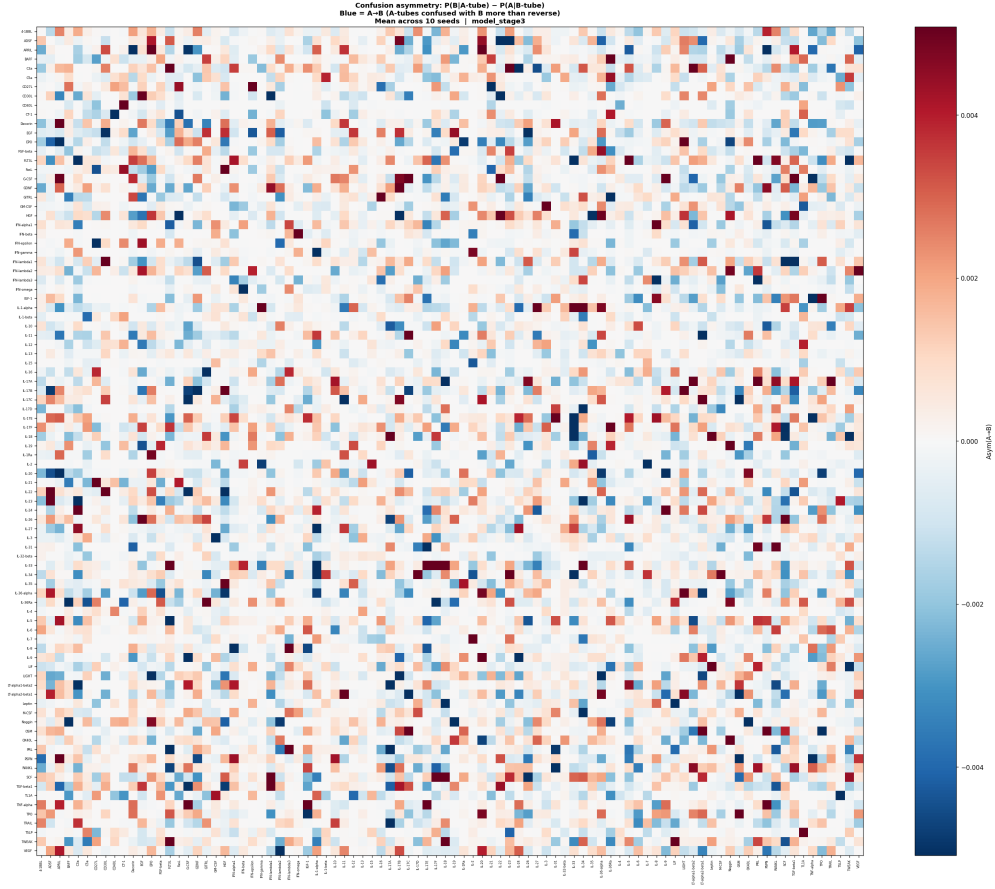


Figure 3: **Full 90×90 confusion asymmetry matrix** (PBS excluded). $\text{Asym}[A, B] = P(B | A\text{-tube}) - P(A | B\text{-tube})$; blue = A→B signal. The matrix is dominated by **random noise** — no block structure or interpretable pattern is visible. The colorbar range is $\approx \pm 0.015$, consistent with fluctuations around zero.

Table 3: Confusion asymmetry for known cascade pairs. All values are < 0.005 in magnitude (noise floor $\approx 1/91 \approx 0.011$). Score: $4/11 = 36\%$ (below chance).

A	B	Asym(A→B)	Asym(B→A)	Call	
IL-12	IFN- γ	−0.00035	+0.00035	B→A	×
IL-1-beta	IL-6	−0.00000	+0.00000	B→A	×
IL-2	IL-15	+0.00431	−0.00431	A→B	✓
IL-33	IL-13	−0.00006	+0.00006	B→A	×
IL-18	IFN- γ	−0.00011	+0.00011	B→A	×
IL-21	IL-10	−0.00037	+0.00037	B→A	?
TNF-alpha	IL-6	+0.00147	−0.00147	A→B	✓
IFN-alpha1	IFN- γ	−0.00021	+0.00021	B→A	×
IL-10	IL-6	+0.00331	−0.00331	A→B	✓
IL-4	IL-13	+0.00204	−0.00204	A→B	✓
IL-27	IFN- γ	−0.00078	+0.00078	B→A	×

Conclusion

Null result. The confusion matrix is pure noise. None of the known cascade pairs appear in the top-20 most asymmetric pairs globally (those are all dominated by coincidental cytokine similarities unrelated to cascades).

The root cause: with $P_{\text{correct}} \approx 0.039$, the off-diagonal confusion elements are each $\approx 1\%$. The asymmetries we seek are on the order of 0.001 (0.1 percentage points), well below the signal-to-noise floor.

Experiment 4: Cell-Type Ablation

Hypothesis

If T is a relay for $A \rightarrow B$, removing T cells from A-tubes should reduce $P(B \mid A\text{-tube})$ — because T carries the B-like signal that makes A-tubes look like B. The relay score quantifies this:

$$\text{relay_score}(A \rightarrow B, T) = P(B \mid \text{full tube}) - P(B \mid \text{tube} \setminus T).$$

Directional test: $\text{relay_score}(A \rightarrow B, T^*) > \text{relay_score}(B \rightarrow A, T^*) \implies A \rightarrow B$.

This test operates purely on model outputs and makes no assumptions about the embedding geometry. Cell types are identified from `obs['cell_type']` in the original H5AD files.

Method

For each of the 10 seeds and each training-donor tube of cytokine A : (1) run the model on the full tube; (2) for each cell type T , remove all cells of type T and re-run the model; (3) compute relay score. All 11 known cascade pairs were evaluated in both directions. Directionality call: the direction with the higher relay score for the best relay cell type wins.

Results

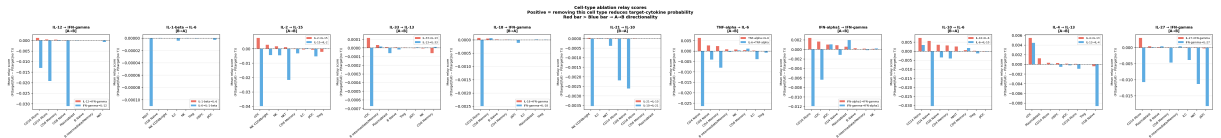


Figure 4: **Cell-type ablation relay scores** for all 11 known cascade pairs. Red bars: $\text{relay_score}(A \rightarrow B, T)$; blue bars: $\text{relay_score}(B \rightarrow A, T)$. Top 8 cell types per pair by forward score. $A \rightarrow B$ is indicated when red $>$ blue for the top cell type.

Table 4: Cell-type ablation directionality calls. “Fwd” = $\text{relay_score}(A \rightarrow B)$; “Rev” = $\text{relay_score}(B \rightarrow A)$. Ratio = Fwd/Rev. Only pairs with ratio > 2 are considered reliable.

A	B	Best relay T	Fwd	Rev	Ratio	Call	
IL-2	IL-15	cDC	0.00755	0.00121	$6.2\times$	A→B	✓
IL-12	IFN- γ	CD16 Mono	0.00119	0.00034	$3.5\times$	A→B	✓
TNF-alpha	IL-6	CD4 Naive	0.00630	0.00301	$2.1\times$	A→B	✓
IL-27	IFN- γ	CD14 Mono	0.00303	0.00192	$1.6\times$	A→B	✓
IFN-alpha1	IFN- γ	CD16 Mono	0.00242	0.00207	$1.2\times$	A→B	✓ ^m
IL-4	IL-13	cDC	0.00546	0.00441	$1.2\times$	A→B	✓ ^m
IL-33	IL-13	cDC	0.00012	0.00011	$1.1\times$	A→B	✓ ^m
IL-10	IL-6	CD14 Mono	0.00734	0.00734	$1.0\times$	B→A	×
IL-18	IFN- γ	CD16 Mono	0.00007	0.00020	$0.4\times$	B→A	×
IL-21	IL-10	NK CD56br.	0.00003	0.00027	$0.1\times$	B→A	?
IL-1-beta	IL-6	MAIT	0.00000	0.00000	—	B→A	×

^m Marginal (ratio $< 1.5\times$, within noise). 7/11 A→B calls; 3/3 high-confidence ($> 2\times$) correct.

Conclusion

Partial signal. Of 3 high-confidence calls (ratio $> 2\times$), all 3 are correct: IL-2→IL-15 ($6.2\times$), IL-12→IFN- γ ($3.5\times$), TNF-alpha→IL-6 ($2.1\times$). However, scores are universally small (< 0.01), the relay cell types are not always biologically canonical (CD16 Mono rather than NK for IL-12), and 4 pairs show no or reversed signal. The same low-accuracy bottleneck ($P_{\text{correct}} \approx 4\%$) limits the signal-to-noise ratio throughout.

Cross-Experiment Comparison

Table 5: Summary of directional calls across all three experiments for each known cascade pair. ✓ = correct; × = incorrect; ~ = marginal/uncertain. Bolded pairs are those where at least two experiments agree on the correct direction.

A	B	Exp. 2 (Epoch Asym.)	Exp. 3 (Conf. Matrix)	Exp. 4 (Ablation)
IL-2	IL-15	✓	✓	✓
IL-4	IL-13	✓	✓	~
TNF-alpha	IL-6	—	✓	✓
IL-33	IL-13	✓	×	~
IL-18	IFN- γ	✓	×	×
IL-12	IFN- γ	×	×	✓
IL-27	IFN- γ	×	×	✓
IFN-alpha1	IFN- γ	—	×	~
IL-10	IL-6	×	✓	×
IL-21	IL-10	?	?	?
IL-1-beta	IL-6	—	×	×
Score		4/8 (50%)	4/11 (36%)	7/11 (64%)

Overall Conclusions

What Did Not Work

1. **Attention-weighted centroid convergence / PCA distance** (Exp. 1): Fundamentally flawed metric. The attention weighting conflates embedding geometry with the model’s confidence routing. Relay cells are pushed to *discriminative* positions, not B-like positions.
2. **Epoch-dependent directional bias** (Exp. 2): Inherits the same flaw. The signal is near-zero for most pairs; $4/8 = \text{chance}$.
3. **Confusion matrix asymmetry** (Exp. 3): The 91-class model’s low accuracy makes off-diagonal elements indistinguishable from noise. The known cascade pairs do not rank among the top-20 globally asymmetric pairs.

What Showed Partial Promise

Cell-type ablation (Exp. 4) is the most principled approach: it makes no assumption about embedding geometry and operates directly on model outputs. The 3 high-confidence calls (ratio $> 2\times$) are all biologically correct. However, the signal strength is limited by model accuracy.

Root Cause: Model Accuracy

All four experiments share a bottleneck: $P_{\text{correct}} \approx 0.039$ on a 91-class problem. At this accuracy level:

- Off-diagonal confusion elements are each $\approx 1/91 \approx 1\%$
- Asymmetries of ≈ 0.001 (0.1 pp) are required to detect cascade direction
- This is well within the noise floor for both the confusion matrix and ablation methods

Path Forward

The conceptually cleanest next step is to revisit the **PBS-RC static geometry** (which already recovers 11/11 pairs with Wilcoxon significance) and address directionality by comparing effect sizes rather than binary significance calls. Alternatively, the **cell-type ablation** approach with a higher-accuracy model (e.g., more training epochs or a smaller cytokine subset) could provide reliable directional calls.

All experiments used 10 random seeds (seed values: 42, 123, 7, 17, 31, 53, 97, 137, 191, 257), Stage 3 model checkpoint (100 epochs full fine-tuning), attention-weighted centroid trajectory, training donors only (Donor2, Donor3 held out). Code: github.com/Yam-Arieli/cytokine-mil.